



Enterprise Architecture: How to Design Models & Diagrams

About Me

- I'm a career Management Consultant for over 20 years
- I specialise in IT Strategy, Cyber Security, Enterprise Architecture and Business Process Management
- Keynote speaker across Australia and South-East Asia
- I've worked alongside many Enterprise Architects over the years and spent time as a Chief Architect myself for a major bank
- I've been an Enterprise Architect trusted advisor to many CIOs representing a variety of industries
- I'm still learning though!

Disclaimer

All information and views presented in this course are my own and of my own research and do not present the views of others or information used by other organisations. Any similarity is purely coincidental.

Enterprise Architecture: how

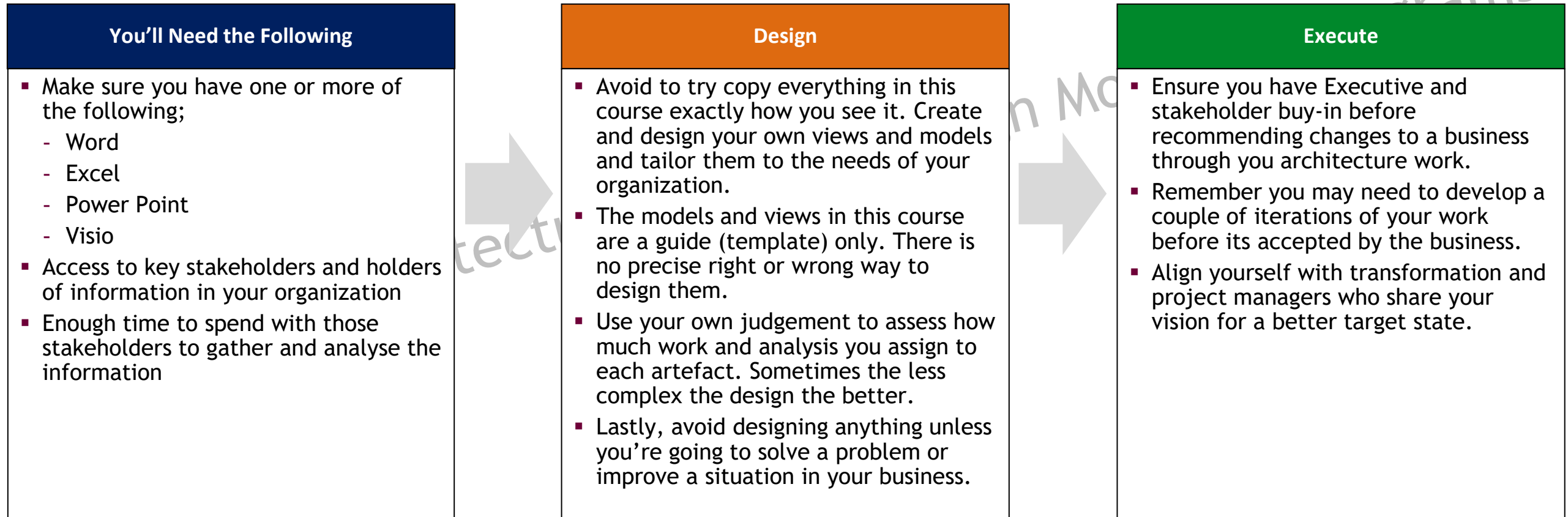
By the end of this course, you should have achieved the following learning outcomes

- Understand Enterprise Architecture and how its used to improve IT and business in an organisation
- Learn about some of the key Enterprise Architecture frameworks
- Learn about the several types of architecture views you can design
- Learn first hand how to design architecture tools and templates you can apply immediately



How to Use the Information in this Course

This course is to help you design your own Enterprise Architecture artefacts





What is Enterprise Architecture?

What is Enterprise Architecture?

1. **Enterprise Architecture is the practice and activity of aligning people, process, governance and technology to achieve the most optimal outcomes for the organization.**

The advantage of effective Enterprise Architecture is...

1

More efficient IT operation

- Lower software development, support and maintenance costs
- Increased portability of applications
- Improved interoperability and easier system and network management
- Improved ability to address critical enterprise-wide issues like security
- Easier upgrade and change of system components

2

Better return on existing investment, reduced risk for future investment

- Reduced complexity in IT infrastructure
- Maximum return on investment in existing IT infrastructure
- The flexibility to make, buy or out-source IT solutions
- Reduced risk overall in new investment, and the costs of IT ownership

3

Faster, simpler and cheaper procurement

- Buying decisions are simpler - available information governing procurement
- Procurement process is faster
- Ability to procure heterogenous, multi-vendor open systems

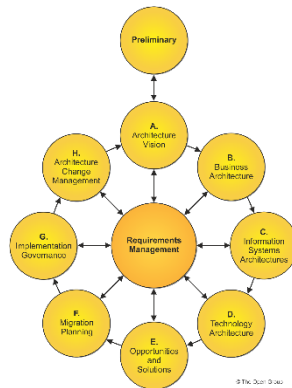


Overview of Enterprise Architecture Frameworks

The following Architecture Frameworks are the most common with each having their pros and cons

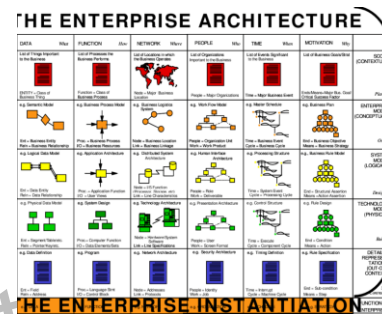
General Frameworks

TOGAF



- World's most used Enterprise Architecture framework
- Covers all parts of an organisation and provides a change management framework
- Can be used for both commercial and government organisations
- Easy to understand and execute

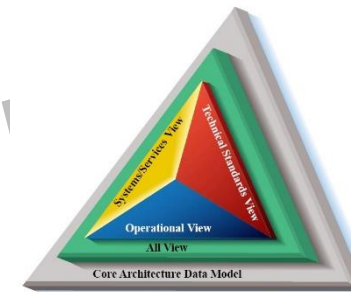
Zachman Framework



- Popular in the late 90s and 2000s but has since disappeared as a result of Zachman's retirement
- Provides a simple framework for artefact development but not a comprehensive change management framework such as TOGAF
- Models and views are also reflected in TOGAF and has similar objectives

Industry Specific Frameworks

DoDAF



- Highly technical architecture framework designed for a Defense organization
- Focus is on integrating operations, systems and technology in a highly secure environment
- Models and views reflect the artefacts in TOGAF and therefore has some overlap

BIAN



- Industry specific architecture framework with a focus on banking
- Provides a reference architecture for specific banking functions - such as payments, transactions, deposits etc
- Not as widely used or applied as TOGAF and most organizations use the BIAN reference architecture as part of a TOGAF deployment



Introduction to Enterprise Architecture Tools and Templates

What Are They Used For?

Validate an Existing State of an Organization

- A way to get consensus among stakeholders and a way to agree on the current state of a design or process.
- Creates a single source of truth and point in time.
- Generates corporate knowledge about the organization and how it utilises resources to deliver products and services to customers.

Create and Test a Future State of an Organization

- Build a means to change the way an organization can optimise its services. Allows analysts and staff across the organization to test certain hypothesis without committing too many resources or cost.
- Provides a way for stakeholders to agree on what the future state should look like before transformation commences.
- Provides project managers and transformation agents with the blueprint before actually commencing change work.

Identify Gaps in the way an Organization Delivers Services and Products to Customers

- Quickly pinpoints the processes and services missing in an organization. Provides clarity to stakeholders as to what is needed within their business to make it more efficient.
- Provides the ability for Enterprise Architects to optimise those products or services - such as reduce time and cost.

What Types of Architecture Artefacts are There?

Type	Explanation
Framework	A high-level and overarching model that provides guidance on how to apply a set of standards and best practices. On its own it usually cannot be used to achieve an outcome. However, a framework is designed to describe a set of rules and practices that - when used - will lead to a beneficial outcome.
Model	A dynamic image that interconnects multiple data points, subjects or objects. Models are iterative in their nature in that when developed or changed, they can have cascading effects across the rest of the representation. They are therefore considered live working documents and are often created using dynamic tools.
View	According to TOGAF, an architecture view is a representation of an overall architecture that is meaningful to one of more stakeholders.
Diagram	Similar to a model, a diagram is usually a flat file image or representation of an architecture artefact that when created, will not have cascading effects on other artefacts. Diagrams are therefore static and usually stand-alone images or representations.



Business Architecture: Tools and Templates

What is Business Architecture?



1

Business Architecture

- Defines the business strategy, governance, organization and key business processes.

2

Data Architecture

- Describes the structure of an organization's logical and physical data assets and data management resources.



3

Infrastructure Architecture

- Describes the logical software and hardware capabilities that are required to support the deployment of business, data and application services. Includes; IT infrastructure, middleware, networks and communications.

4

Application Architecture

- Provides a blueprint for the individual applications systems to be deployed, and their interactions, and their relationships to the core business processes of the organization.





Business Strategy View

Business Strategy View

Format Type: Usually Word or PowerPoint format



Description

A strategic document which describes the future vision of the organisation, how it will create competitive advantage, and growth in the market place over the next two years.



Information Required

- Annual Report
- Consumer Intelligence
- Competitor Intelligence
- Future Trends Analysis
- Organization size and existing capabilities
- Organizational resource effectiveness



Intended Audience

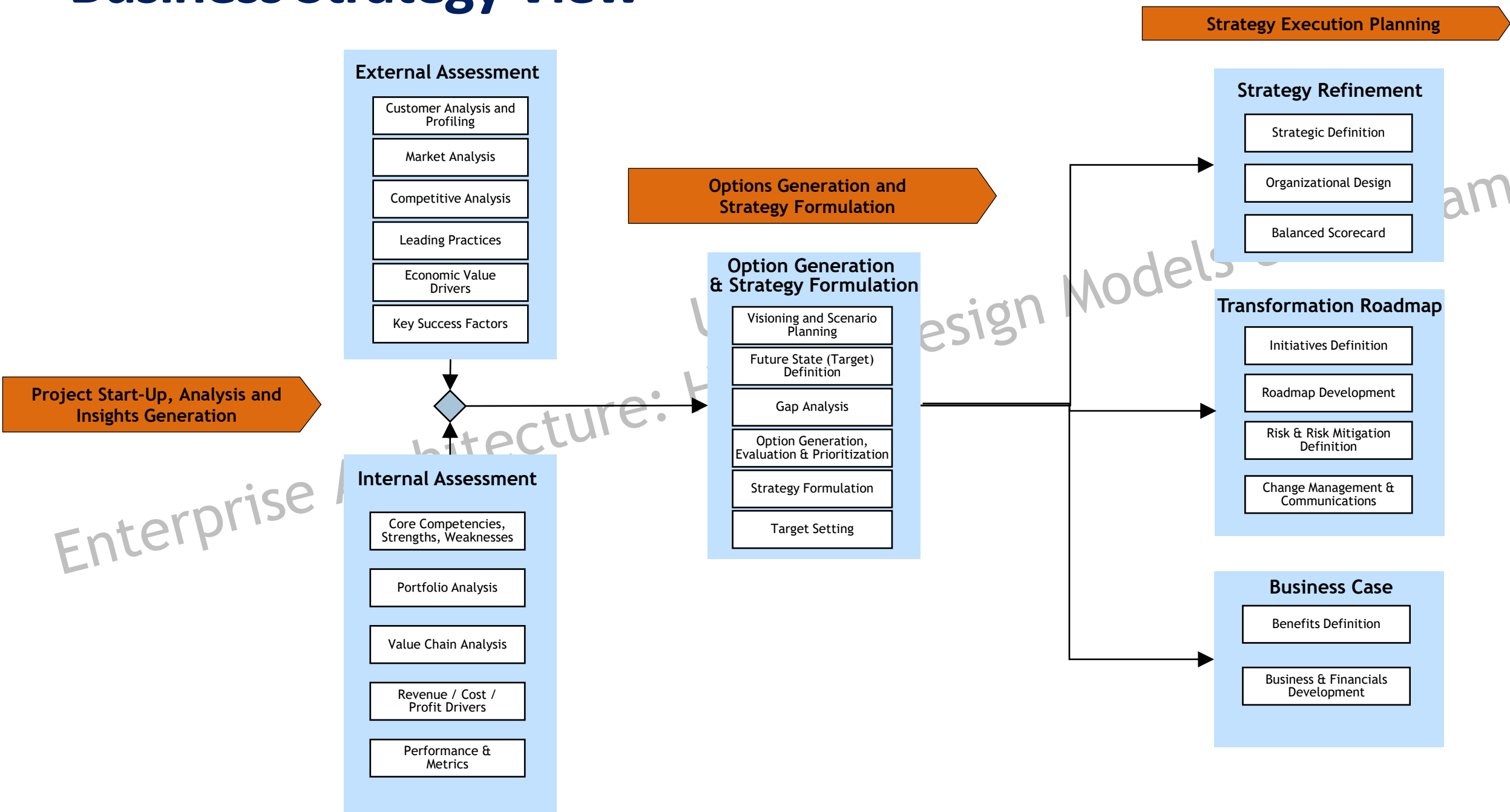
- The Board
- CEOs
- C-Suite Executives
- Can be either made public or be a private document only accessible to certain stakeholders



Key Benefits

- Clearly articulates the CEOs direction for the company.
- Allows IT and other portfolios to align their planning around the strategy.
- Describes how the organization will become a market leader and what is required to achieve that ambition.

Business Strategy View



I Organization View

Organization View

Format Type: Usually Visio or PowerPoint Diagram



Description

A high-level document which describes how the organization is structured according to portfolio, team and role. It contains role and team descriptions but not individual names.



Information Required

- HR Staff roll or,
- Internal staff directory



Intended Audience

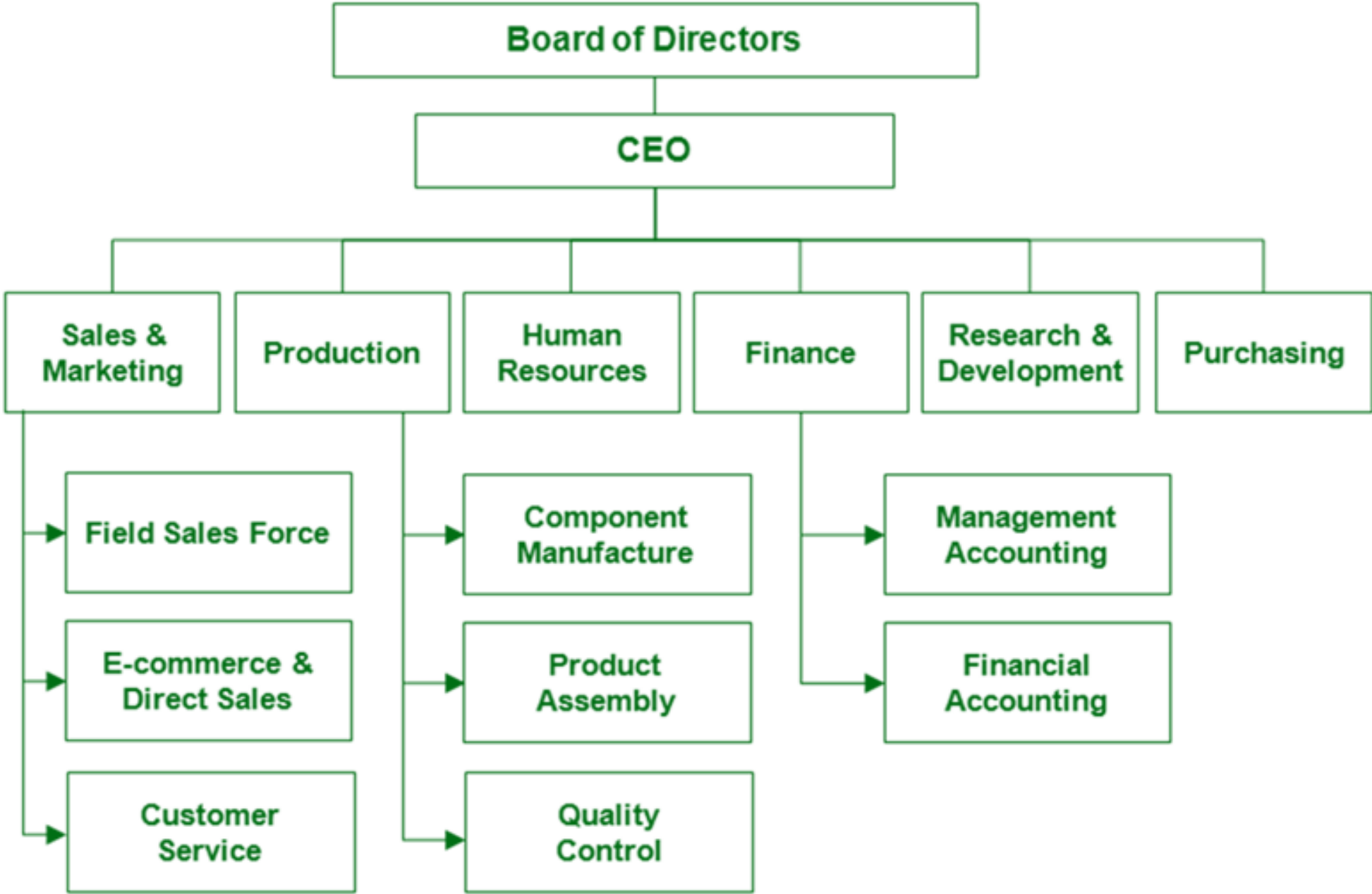
- Everyone within the organization
- Third-party vendors such as consultants or outsourced project managers



Key Benefits

- Provides key insight into the size and/or complexity of the organisation
- Gives IT Architects the ability to assess how best to align and optimise IT services
- Shows Business Architects and leaders if there are overlaps or duplication of work of functions across the organization

Organization View





Operational Model

Operational Model

Format Type: Real life picture or graphic image



Description

Provides a real life visual view – via an image – of what an operational scenario would look like once designed and implemented correctly.



Information Required

- To develop this model you'll need to workshop the target state with business stakeholders.
- You'll may also need an expert Photoshop wizard to help you design the final image.



Intended Audience

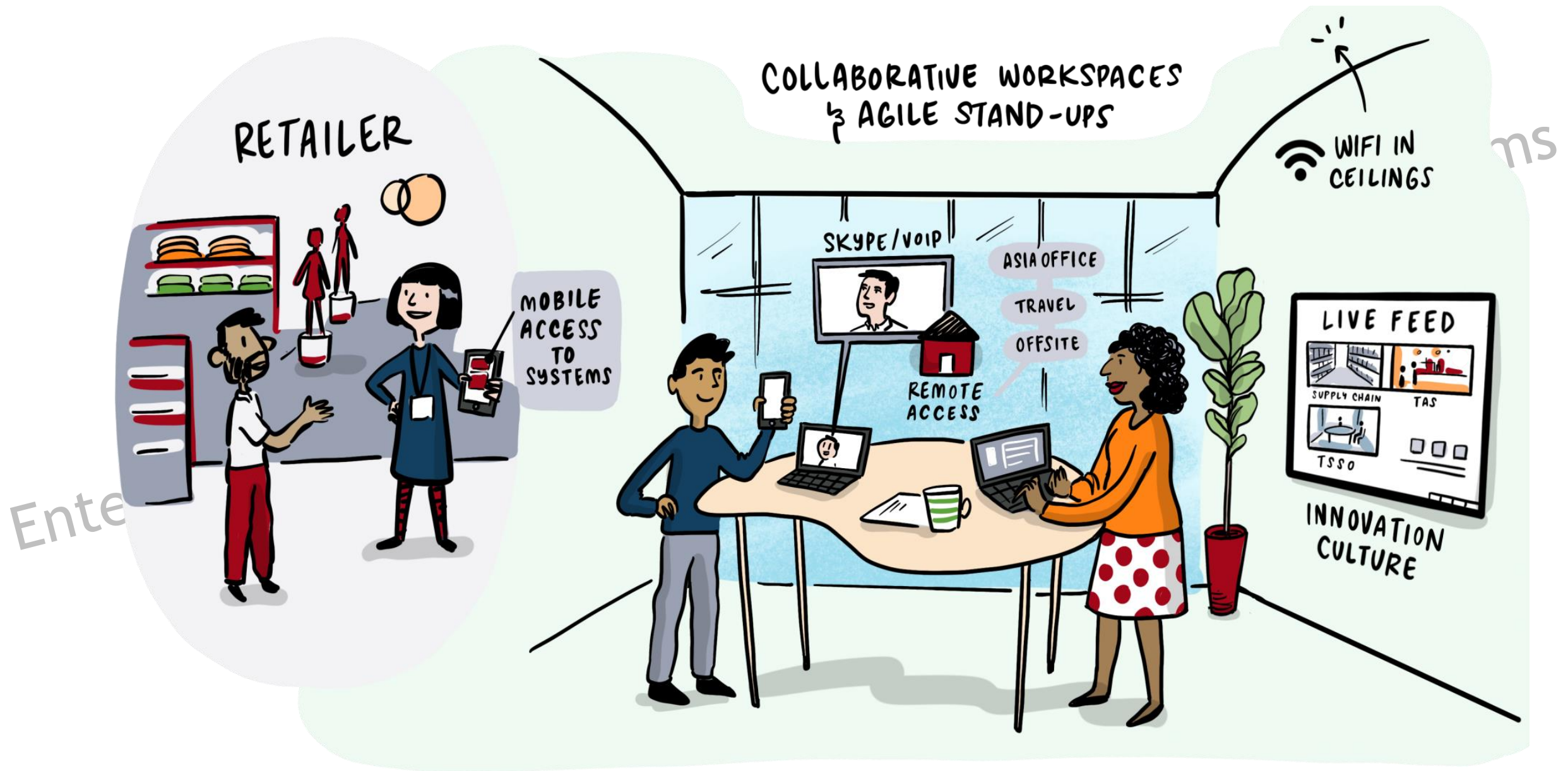
- Business Executives
- Business operations or management personnel
- IT Architects
- Security Architects
- Third-party suppliers
- Marketing teams



Key Benefits

- Helps provide all stakeholders with a visual image of what a particular scenario could look like. It therefore allows stakeholders to debate the target and look for ways to improve it without have to commit much budget or time overhead.

Operational Model





Business Service View

Business Service View

Format Type: PowerPoint or Visio Diagram



Description

A diagram that depicts the entire list of services an organization delivers and groups those services by function.



Information Required

- The organization chart is the best place to start. Following that, attempt to develop the organization's list of services yourself using industry frameworks such as BIAN or APQC.



Intended Audience

- CEO and Business Executives.
- IT and service delivery teams.
- Enterprise Architects.

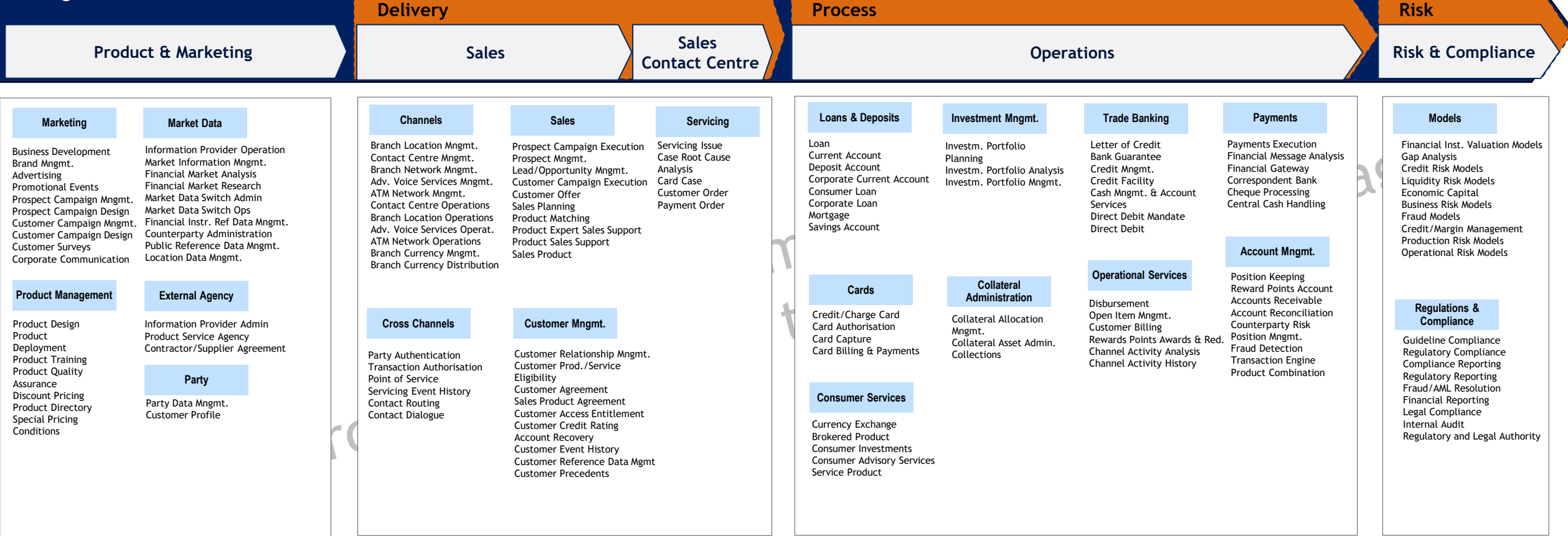


Key Benefits

- Describes exactly what the organizations does do versus what it doesn't do.
- Allows Enterprise and IT Architects to assign IT services and assess which parts of the organization are under resourced in terms of IT need.
- Can also show those parts of the business which are under or over resourced in terms of people.

Business Service View

Banking Services



Enterprise and Enabling Support Services



I Persona View

Persona View

Format Type: Word document or PowerPoint



Description

A series of analytical charts that describe the different type of people that would use services or products for an organization. These could either be customers or the organizations own internal staff.



Information Required

- Marketing and customer data.
- HR organization demographic data.
- Market segment scan data or anything that provides customer analytics and customer/ market segmentation.



Intended Audience

- CEO and Business Executives.
- Sales and marketing teams.
- IT and service delivery teams.
- Security Architects.



Key Benefits

- Helps Enterprise Architects exactly align and match products or services to the needs of customers or staff – and draw traceability between the two.

Persona View



The
Indulger

“ It is important that I frequently get my dosage of fun...”

Profile

Name: Robert
Gender: Male
Age: 32

Occupation: Engineer
Tech Savvy

Likes

- Joyful experiences
- Having fun in regular intervals and frequently

Attitude

- Is seeking out a secure and successful career
- Values pleasurable experiences over anything else
- Mobility is key
- Eager to explore and possess all kinds of additional benefits and values in every situation
- Looks at the long term goal rather than just doing things for short term gains
- Wants to feel special, part of the ‘VIP’ group
- Gets reassurance from others telling him he’s successful

Motivations

- Eminence: Inclined towards posting on social media to gain recognition from other of his knowledge / expertise. Knowing that he is capable of helping or influencing others gives him great satisfaction.
- Fun-ness: Regular indulgences are necessary to keep himself recharged; leisure entertainment, travels, social media, games etc
- Pleasurable Experience: Wouldn’t mind paying more to enjoy better services without limitations.

Target Context

- Likes to do retail shopping online
- Will rarely visit a bricks a mortar store
- Tracks usage (spending, balances etc) via website or app
- Wants to accumulate points
- Checks for rewards + promotions

I Process Catalogue

Process Catalogue (Process Hierarchy)

Format Type: Word document or PowerPoint



Description

A formal list, in a database format that describes all the processes that an organization delivers either internally or external to its customers. It groups these processes by groups and categories.



Information Required

- You will need the completed version of the *Business Services View* to commence working on a complete Process Catalogue.



Intended Audience

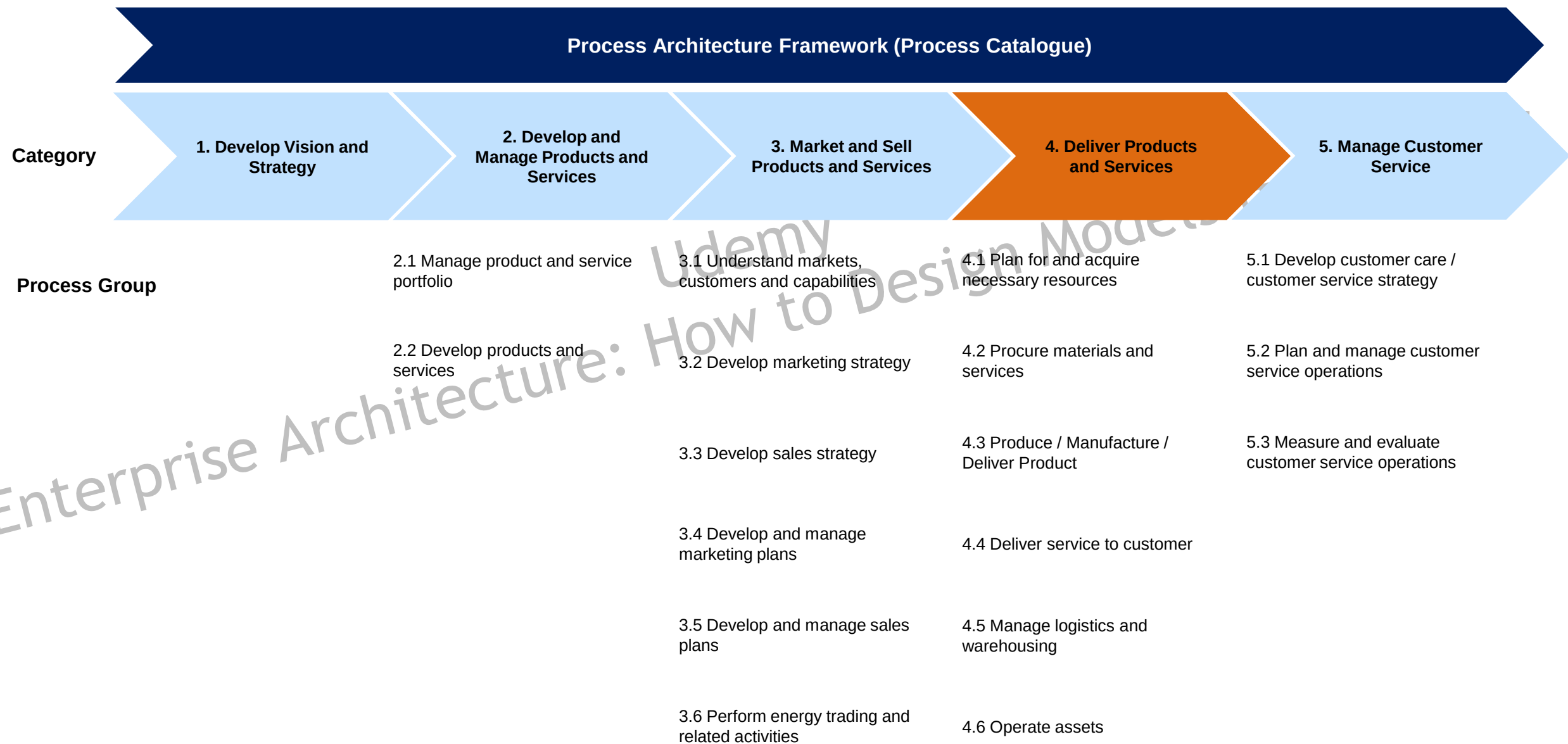
- Business Analysts
- Enterprise Architects
- Project Managers
- Transformation and Change Managers
- Lean and Six Sigma Practitioners



Key Benefits

- Helps quickly identify what processes and organization delivers to customers.
- Creates a formal method for capturing and managing processes.
- Houses all process models which are then used for process optimization analysis activities.

Process Catalogue



I Process Model

Process Model

Format Type: PowerPoint or Visio



Description

A model that describes how a process is executed within an organization.



Information Required

- Ideally, you would want to complete the *Process Catalogue* first before assigning and designing process models.
- You will also need business stakeholders and maybe customers to help you define the process itself.



Intended Audience

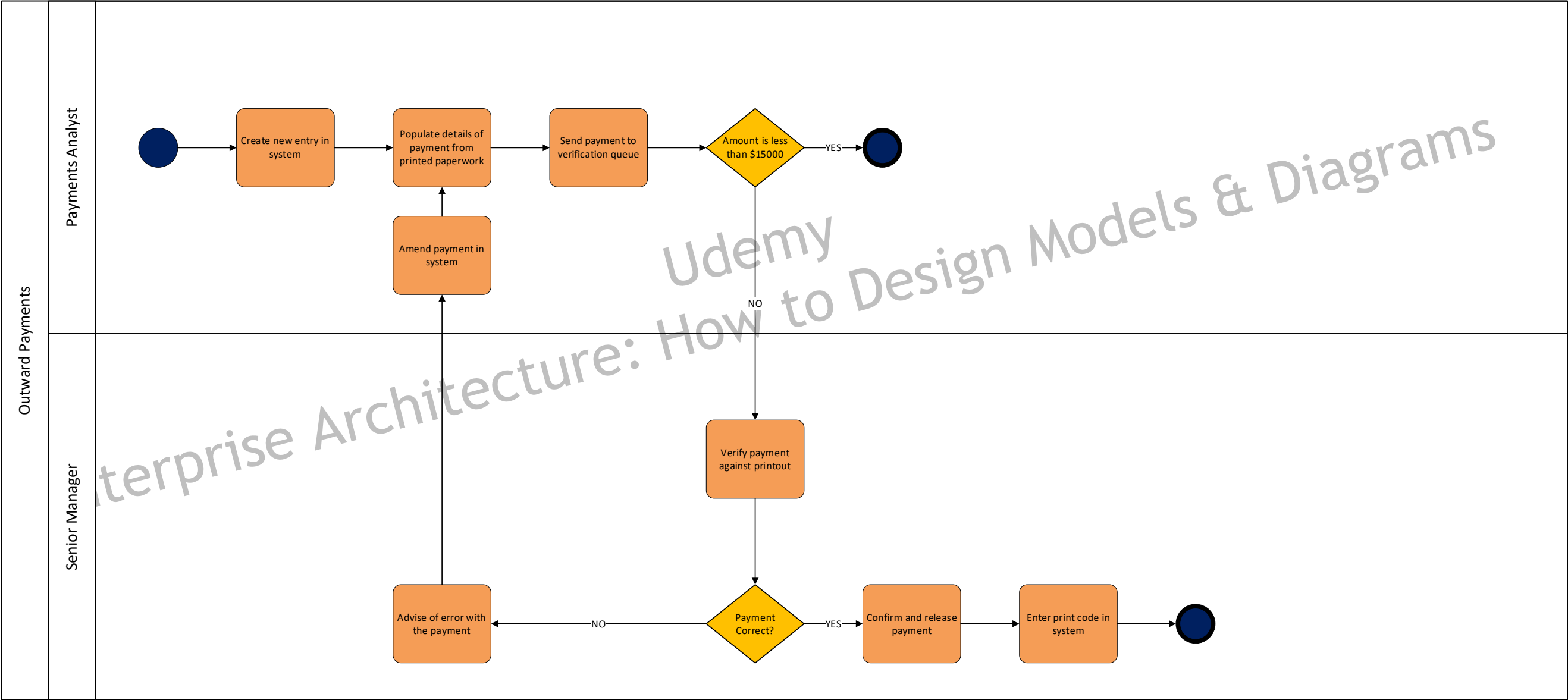
- Business Analysts
- Enterprise Architects
- Project Managers
- Transformation and Change Managers
- Lean and Six Sigma Practitioners



Key Benefits

- Minimizes debate among business users as to how a process is executed.
- Allows analysts to look for process improvements to help save time and cost within an organization.

Process View





Business Service and Information Diagram

Business Service and Information Diagram

Format Type: Excel Spreadsheet, PowerPoint or Visio



Description

Shows the information required to support one or more business services.



Information Required

- Recommend you go to the business unit that delivers the service and capture all information they use in an Excel spreadsheet before moving to building it in a diagram.



Intended Audience

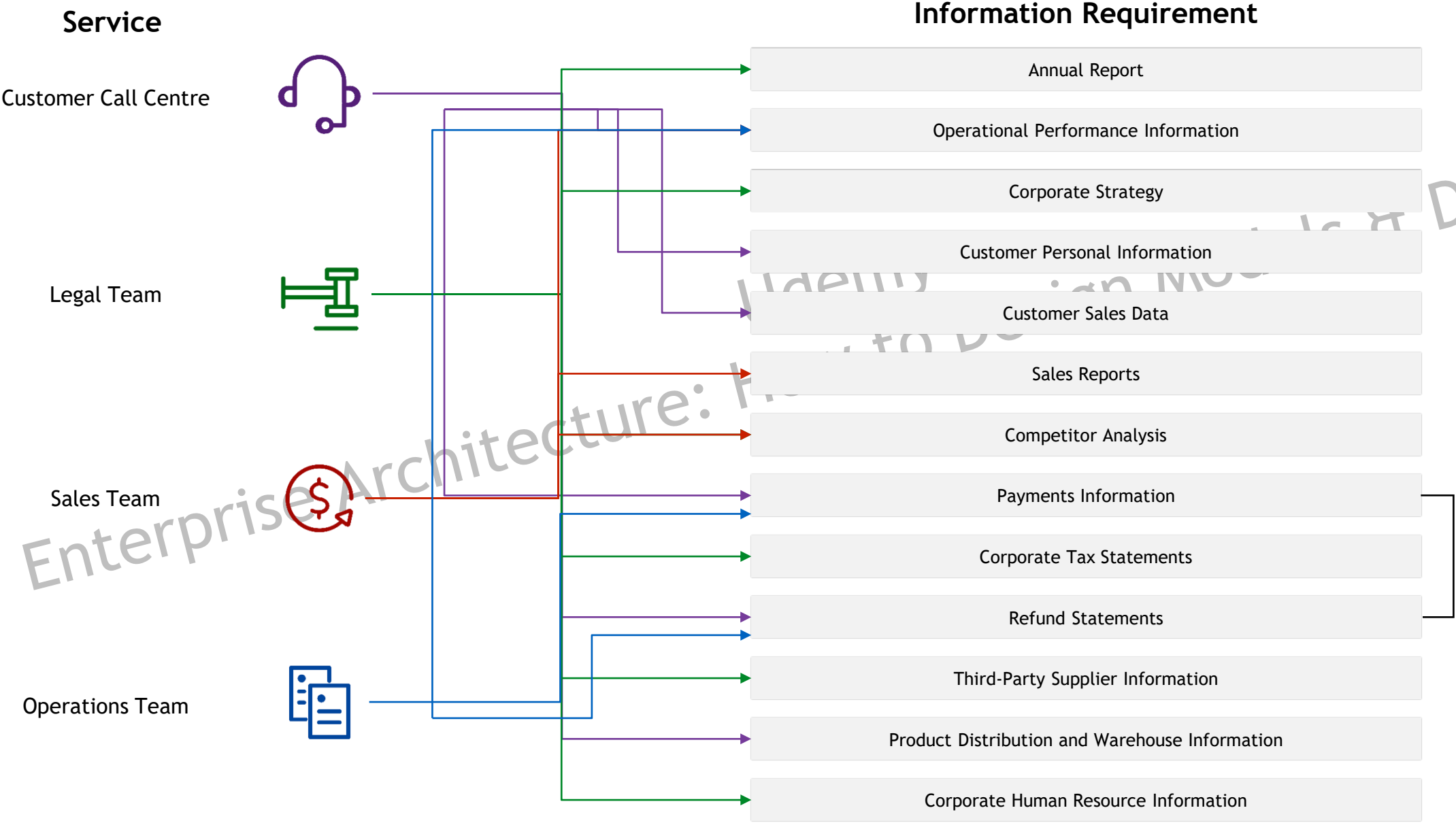
- Business Analysts
- Enterprise Architects
- Business leaders and Executives
- Data Analysts
- IT Analysts



Key Benefits

- This diagram confirms the information a business needs to deliver services as a source of truth.
- Secondly, it describes which IT would be needed to support those information requirements and ensures the right level of resources and capability is given without overspend.

Business Service and Information Diagram





Functional Decomposition Diagram

Functional Decomposition Diagram

Format Type: Excel Spreadsheet



Description

A functional decomposition diagram contains the whole function or project along with all of the necessary sub-tasks needed to complete it. Functional decomposition is a problem-solving tool used in several contexts, from business and industry to computer programming and AI.



Information Required

- Use the *Process Catalogue* and *Business Service View* to help quickly build out this diagram.
- You'll also probably need to spend a lot of time with that function or business unit to capture all sub-tasks.



Intended Audience

- Business Analysts
- Enterprise Architects
- Business leaders and Executives
- Data Analysts
- IT Analysts



Key Benefits

- A very quick way to capture processes before you start modelling them in a Business Process Model.
- Allows you to quickly create, move or delete processes in an inventory style manner.
- Ensures all stakeholders are in agreeance as to what processes and tasks a business unit actually executes.

Functional Decomposition Diagram

[illegible]



Business Locations Catalogue

Business Locations Catalogue

Format Type: Excel Spreadsheet



Description

A spreadsheet that lists the locations all of the organizations offices or branches and their role or function.



Information Required

- This sort of information is usually held by a facilities manager who may sit in your operations group.



Intended Audience

- Business Leaders and Executives
- Business Analysts
- Enterprise Architects
- IT Analysts
- Facilities Managers
- Human Resources Personnel



Key Benefits

- For many Architects, this would be one of the few means to assess how geographically disperse the organization is.
- Allows to identify how to best allocate people and IT services to support those locations.
- Or, for Business Architects, helps assess which locations are performing/under-performing.

Business Locations Catalogue

File

Home

Insert

Page Layout

Formulas

Data

Review

View

Help

Tell me what you want to do

Cut

Copy

Format Painter

Calibri

10

A

A

B

I

U

A

Wrap Text

General

Conditional Formatting

Format as Table

Normal 2 2

Normal 3

Normal

Bad

Good

Neutral

Calculation

Check Cell

Explanatory T...

Followed Hyp...

Insert

Delete

Format

AutoSum

Fill

Clear

A Z

Sort &

Find &

Filter

Select

Clipboard

Font

Alignment

Number

Styles

Cells

Editing

X30

A

B

C

D

E

F

G

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I

J

K

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M

N

O

P

Q

R

S

T

U

V

W

X

Y

Z

AA

AB

2

Business Locations Catalogue

4

California Changes / Moves

Region 1		Action	Region 2		Action	Region 3		Action
90201	Bell Gardens		90248	Gardena		90264	Malibu	
90201	Cudahy		90249	Gardena		90265	Malibu	ADD
90202	Bell		90250	Holly Park		90266	Manhattan Beach	
90202	Bell Gardens		90250	Hawthorne		90267	Manhattan Beach	
90209	Beverly Hills		90251	Hawthorne		90270	Bell	
90210	Beverly Hills	REMOVE	90254	Hermosa Beach	REMOVE	90270	Maywood	REMOVE
90211	Beverly Hills		90255	Walnut Park		90272	Pacific Palisades	
90212	Beverly Hills		90255	Huntington Park		90274	Palos Verdes Peninsula	
90213	Beverly Hills		90260	Lawndale		90274	Palos Verdes Estates	
90220	Rancho Dominguez		90261	Lawndale		90274	Rolling Hills	
90220	Compton		90262	Lynwood		90274	Rolling Hills Estates	
			90263	Malibu				

20

New York Changes / Moves

Region 4	Paul Everett	Action	Region 5	Matt Jones	Action	Region 6	Matthew Hallal	Action
11901	Flanders		11960	Remsenburg		13801	McDonough	
11901	Riverhead		11961	Ridge		13802	Maine	
11930	Amagansett		11962	Sagaponack		13803	Marathon	
11931	Aquebogue		11963	Sag Harbor		13804	Masonville	
11932	Bridgehampton		11964	Shelter Island		13806	Meridale	
11933	Calverton		11965	Shelter Island Heights		13807	Milford	
11934	Ctr Moriches		11967	East Yaphank	ADD	13808	Morris	
11934	Center Moriches		11967	Shirley		13809	Mount Upton	
11935	Cutchogue		11968	Southampton		13810	Mount Vision	
11937	East Hampton		11969	Southampton		13811	Newark Valley	
11939	East Marion		11970	South Jamesport		13812	Nichols	
11940	East Moriches		11971	Southold		13813	Nineveh	
11941	Eastport		11972	Speonk	ADD	13814	North Norwich	
11942	East Quogue		11973	Upton				
11944	Greenport		11975	Wainscott				
			11976	Water Mill				

4

California Structure (new)

Region 1		Region 2		Region 3	
90201	Bell Gardens	90248	Gardena	90264	Malibu
90201	Cudahy	90249	Gardena	90265	Malibu
90202	Bell	90250	Holly Park	90266	Manhattan Beach
90202	Bell Gardens	90250	Hawthorne	90267	Manhattan Beach
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90220	Rancho Dominguez	90261	Lawndale	90274	Rolling Hills
90220	Compton	90262	Lynwood	90274	Rolling Hills Estates
		90263	Malibu		

4

New York Structure (new)

Region 4	Paul Everett	Region 5	Matt Jones	Region 6	Matthew Hallal
11901	Flanders	11960	Remsenburg	13801	McDonough
11901	Riverhead	11961	Ridge	13802	Maine
11930	Amagansett	11962	Sagaponack	13803	Marathon
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11934	Ctr Moriches	11967	East Yaphank	13808	Morris
11934	Center Moriches	11967	Shirley	13809	Mount Upton
11935	Cutchogue	11968	Southampton	13810	Mount Vision
11937	East Hampton	11969	Southampton	13811	Newark Valley
11939	East Marion	11970	South Jamesport	13812	Nichols
11940	East Moriches	11971	Southold	13813	Nineveh
11941	Eastport	11972	Speonk	13814	North Norwich
11942	East Quogue	11973	Upton		
11944	Greenport	11975	Wainscott		
		11976	Water Mill		



Business Interaction Matrix

Business Interaction Matrix

Format Type: Can be an Excel Spreadsheet or Table



Description

A spreadsheet or table that lists every business unit in an organization and shows the touch points that unit has with other business units within the same organization.



Information Required

- You can usually develop this view once the *Organization View* has been developed.



Intended Audience

- Business Leaders and Executives
- Business Analysts
- Enterprise Architects
- IT Analysts
- Solution Architects



Key Benefits

- This is one of the best views for showing the usefulness and importance of a business unit and its interactions. It helps IT Architects in particular in making decisions on how IT and resources should be prioritized.
- Shows the 'Centre of Gravity' for an organization which can be critical.

Business Interaction Matrix

	Sales Team	HR Team	IT Team	Marketing Team	Legal Team	Business Operations	Customer Service Team	CEO/ Management Team	Finance/ Accounts Team
Sales Team			X	X			X		X
HR Team			X		X				
IT Team	X	X	X	X	X	X	X	X	X
Marketing Team			X						
Legal Team			X					X	
Business Operations	X		X				X		X
Customer Service Team	X		X	X		X			X
CEO/ Management Team	X	X	X	X	X	X	X	X	X
Finance/ Accounts Team	X		X						

Data Architecture: Tools and Templates

What is Data Architecture?



1

Business Architecture

- Defines the business strategy, governance, organization and key business processes.

2

Data Architecture

- Describes the structure of an organization's logical and physical data assets and data management resources.



3

Infrastructure Architecture

- Describes the logical software and hardware capabilities that are required to support the deployment of business, data and application services. Includes; IT infrastructure, middleware, networks and communications.

4

Application Architecture

- Provides a blueprint for the individual applications systems to be deployed, and their interactions, and their relationships to the core business processes of the organization.





Data Governance Model

Data Governance Model

Format Type: Can be Word, Excel, PowerPoint or Visio



Description

A framework used to help define the organization's approach to data management across several different criterion and attributes.



Information Required

- You will need to interact with the teams from the IT Group, in addition to senior IT Executives to build out each of the policies and artefacts for each component of the framework.



Intended Audience

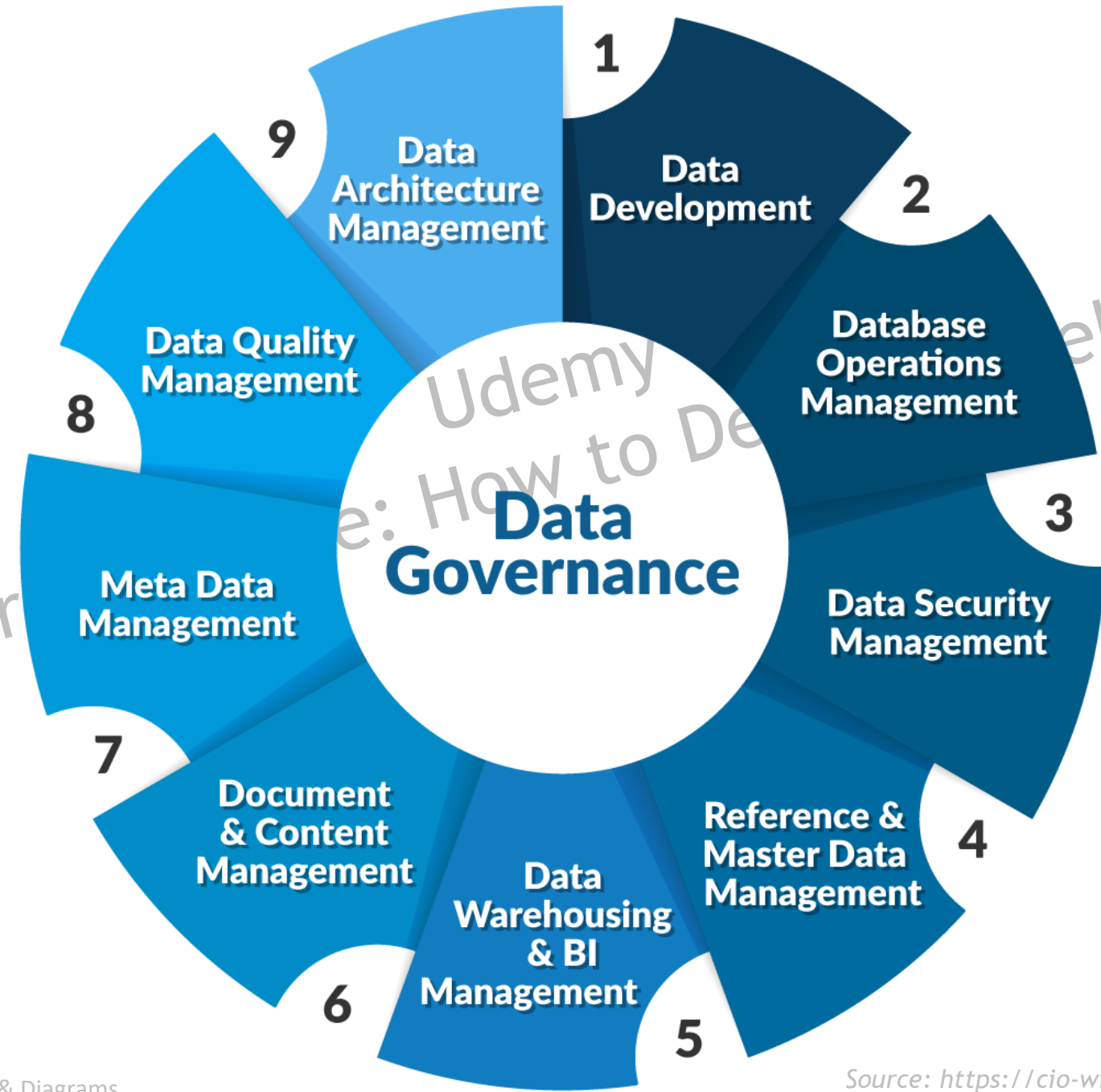
- CIO and Chief Data Officer
- Data Architects and Analysts
- Business Analysts
- Enterprise Architects
- IT Analysts
- Solution Architects
- Project Managers and Consultants



Key Benefits

- Ensures the organization's approach to managing its data is best practice and is well organized, structured and managed to improve the way data supports the business and the organization's customers.

Data Governance Model



Information Requirements View

Information Requirements View

Format Type: Can be Word, Excel, PowerPoint or Visio



Description

A framework that allows Data Architects to identify the data needs of an organization or business unit.



Information Required

- You will need to hold stakeholder workshops with business teams and leaders to identify their information requirements using this framework.



Intended Audience

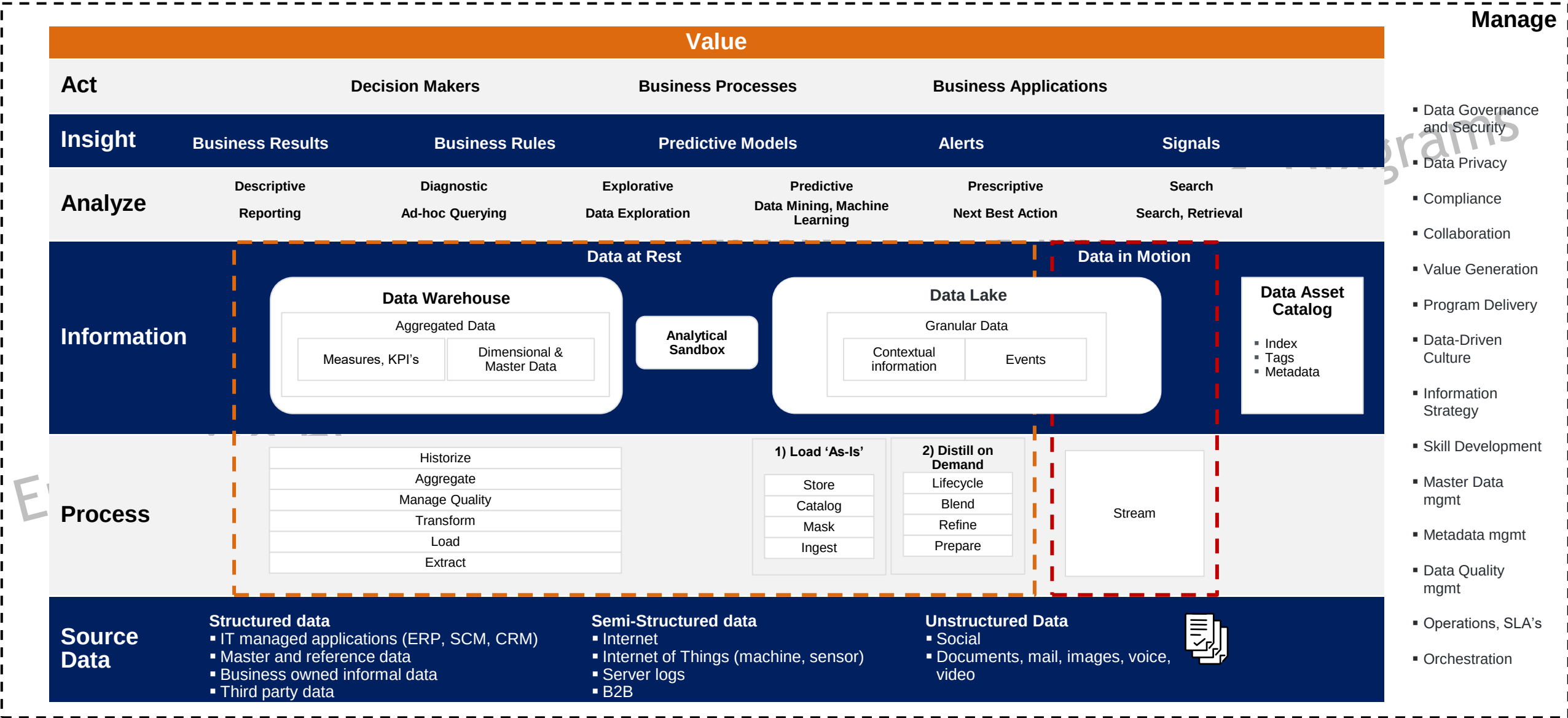
- Business Unit Leaders
- CIO and Chief Data Officer
- Data Architects and Analysts
- Business Analysts
- Enterprise Architects
- IT Analysts
- Solution Architects
- Project Managers and Consultants



Key Benefits

- Provides a structured catalogue and list of all the ways data can add value to an organization or business unit.
- Data Architects can use the framework to help improve the way data is used or managed within an organization.

Information Requirements View





Information Hierarchy View

Information Hierarchy View

Format Type: Can be a Word document, Excel spreadsheet or table



Description

A document that describes how important each policy, document, information or data is in relation to the other.



Information Required

- Usually only business and IT leaders can authorise the importance of each information in terms of how critical it is to an organization.



Intended Audience

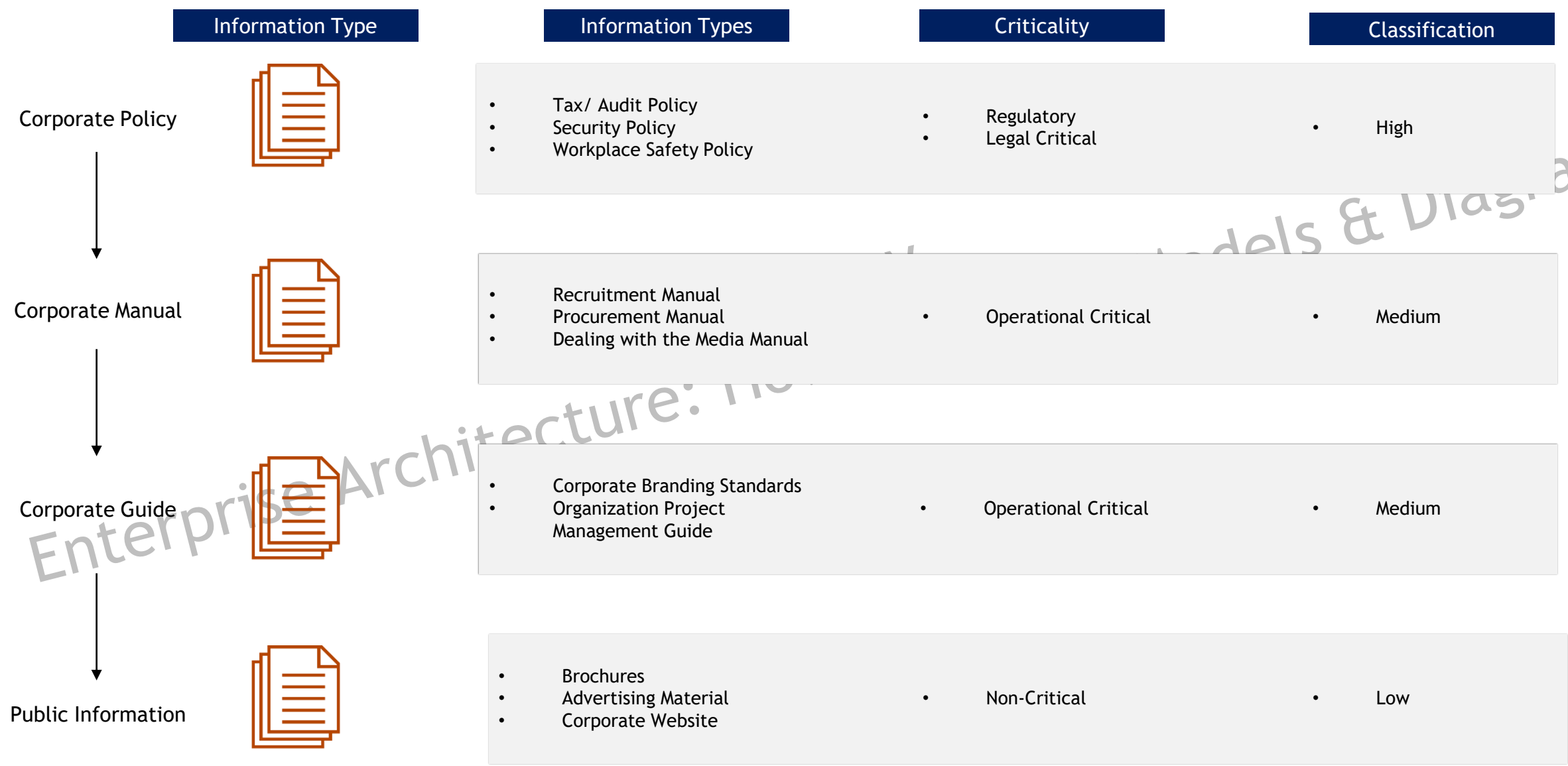
- Business Unit Leaders
- CIO and Chief Data Officer
- Data Architects and Analysts
- Business Analysts
- Enterprise Architects
- IT Analysts
- Solution Architects
- Project Managers and Consultants



Key Benefits

- This type of view is usually only of benefit to data architects and security architects as its helps identify the organizations most critical information/data and also helps prioritize projects.
- Also, the higher the document, the longer the change lifecycle to that document.

Information Hierarchy View



| Logical Data Model

Logical Data Model

Format Type: Visio or PowerPoint diagram



Description

A document that describes how data logically flows from one entity to another and the type of data or information that is used during that event.



Information Required

- Firstly, it is best you develop the *Business Interaction Matrix* as it will help identify data flows.
- Second, you will need to work with each business unit to identify the information and data they each exchange. You can use the *Information Requirements View* to assist.



Intended Audience

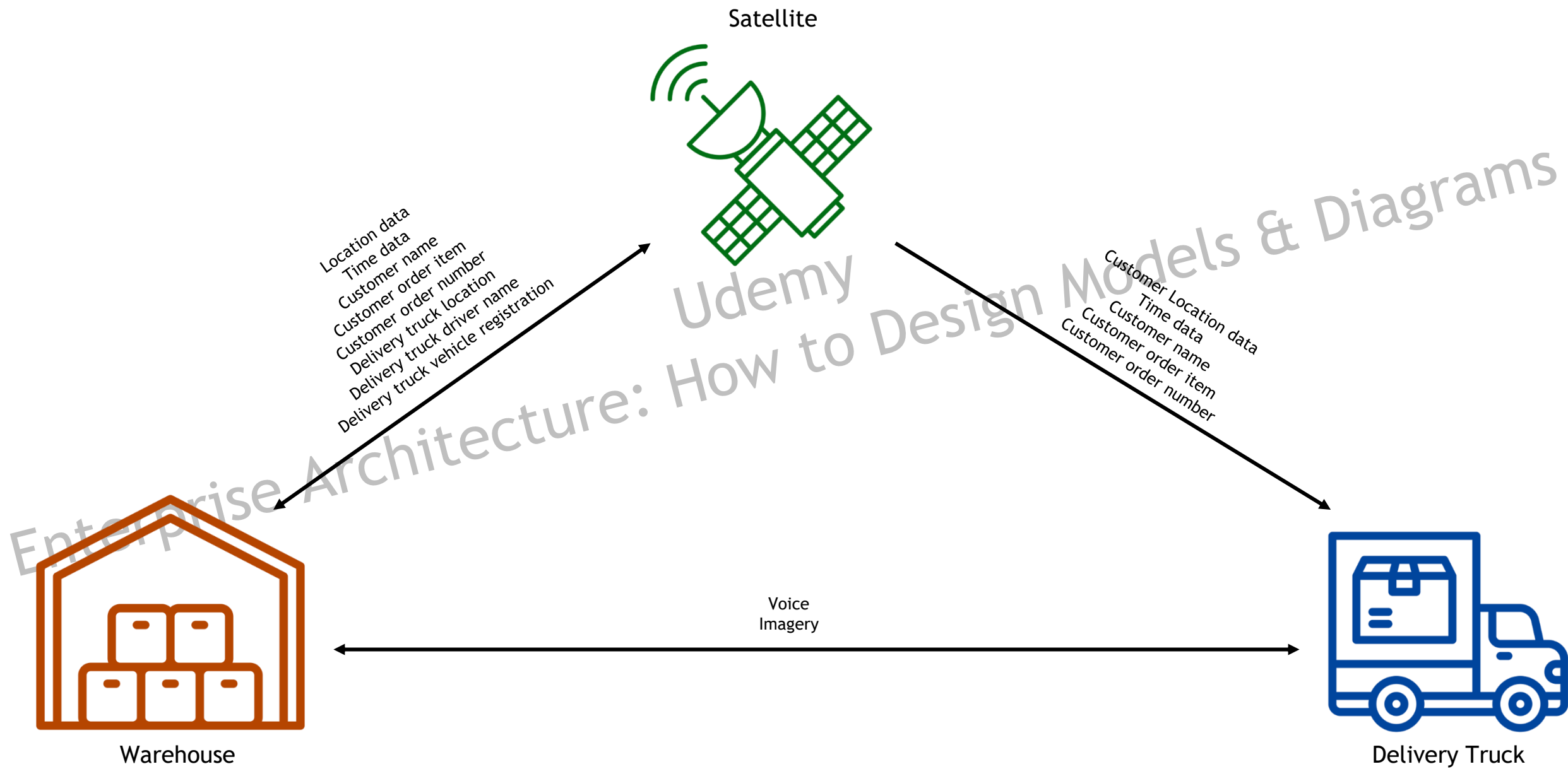
- Business Unit Leaders
- CIO and Chief Data Officer
- Data Architects and Analysts
- Business Analysts
- Enterprise Architects
- IT Analysts
- Solution Architects
- Project Managers and Consultants



Key Benefits

- Forces business teams to think about how they exchange information and data, and whether such exchanges are needed.
- Helps security architects assess how that information or data can be secured.
- Validates the information and data types a business needs in order to function.

Logical Data Model



| Physical Data Model

Physical Data Model

Format Type: Excel Spreadsheet, Table, PowerPoint or Visio



Description

A visual description of data and information that is housed within an IT system or application.



Information Required

It is recommended to speak with the vendor who has built the system or application or the solution architect who manages it.



Intended Audience

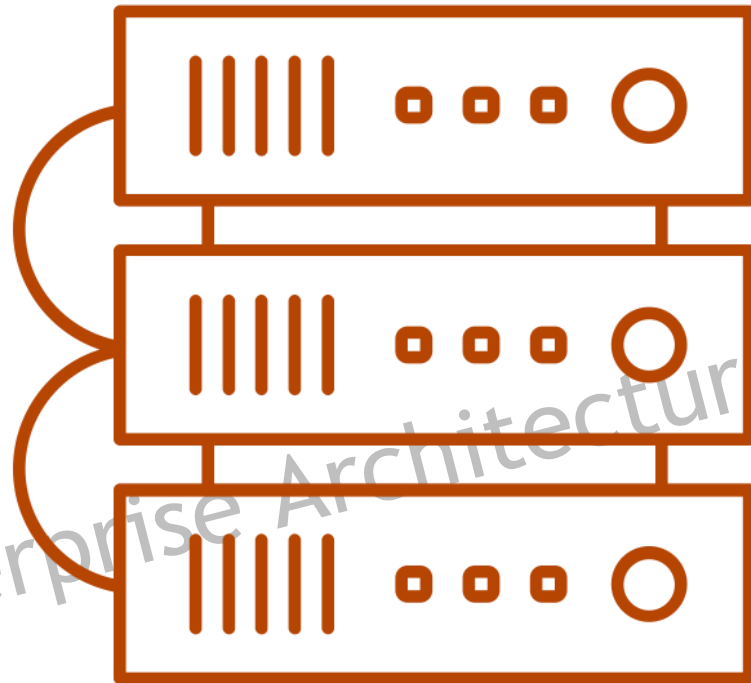
- Security Architects
- Solution Architects
- IT Analysts (Operations, Network, Service Delivery)
- Data Architects
- CIOs



Key Benefits

- This model can provide a critical view of where an organizations data is located, and help identify critical systems.
- This view can also facilitate conversations regarding IT optimization and cyber security.

Physical Data Model



Physical Corporate Server

- Documents
- Imagery
- Audio
- Staff information
- Customer information
- Tax reports
- Financial reports
- Customer surveys
- Corporate website data
- Conference recordings
- Customer receipts
- Payment slips

Physical Exchange Model

Physical Exchange Model

Format Type: PowerPoint or Visio



Description

A model which is used to assess the data that is exchanged between IT systems. Its purpose is to assess interoperability and integration between systems.



Information Required

It is recommended to speak with the vendor who has built the system or application or the solution architect who manages it.



Intended Audience

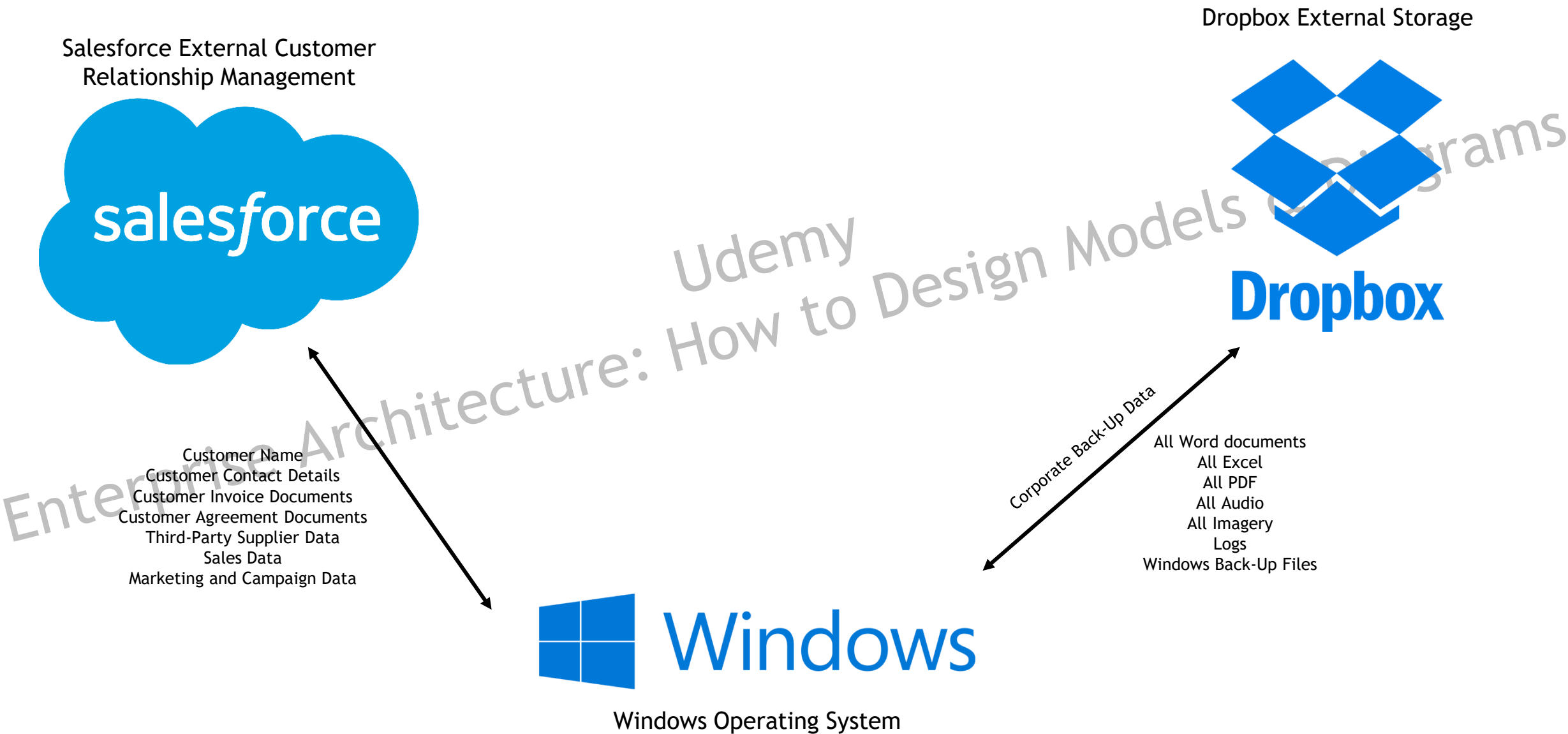
- Security Architects
- Solution Architects
- IT Analysts (Operations, Network, Service Delivery)
- Data Architects
- CIOs
- Project Managers responsible for IT transformation and integration programs



Key Benefits

- This model can provide a critical view of where an organizations data is located, and help identify critical systems.
- Similar to the *Physical Data Model*, this view can also facilitate conversations regarding IT optimization and cyber security.

Physical Exchange Model



Data Security Model

Data Security Model

Format Type: Excel Spreadsheet, table, PowerPoint or Visio



Description

A model that determines and assesses how data types are secured.



Information Required

This type of model can take a long time to complete – particularly if information is absent (such as data governance model). It is recommended to work with the Security Architect on how to gather security-related information for data types.



Intended Audience

- Security Architects
- Solution Architects



Key Benefits

- Provides an exhaustive view on how to classify and secure data.
- Identifies security gaps that may expose data protection vulnerabilities and therefore requires urgent remediation.

Data Security Model



Organization Audit Statement

- Define data owner
- Define data classification
- Define storage location
- Define user access
- Define creation date
- Define archive date
- Define destruction date
- Define back-up location
- Confirm security of back-up location
- Define emergency access
- Define protection at rest (encryption)
- Define protection in transport (encryption)

Data Flow Model

Data Flow Model

Format Type: PowerPoint or Visio



Description

A representation of the flow and exchange of information within a system or as part of a process, and how data is moved from one system to another or one process to another.



Information Required

For this model it is common that the business owners who execute or use the data will have knowledge of data flows within their own business unit. Therefore it is best to hold a workshop with these stakeholders to gather the required information.



Intended Audience

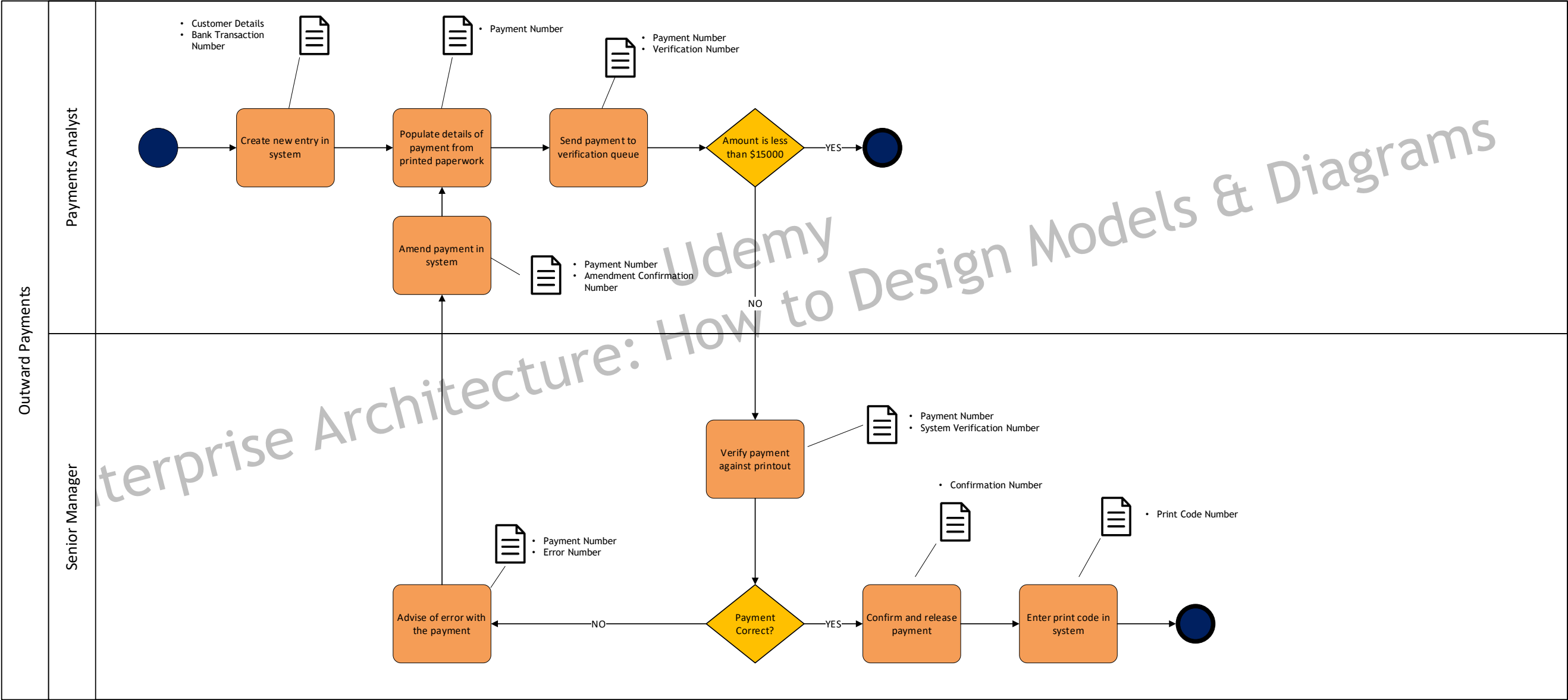
- Business Analysts
- Business Process Architects
- System/ Solution Architects
- Data Architects
- Security Architects
- Project Managers responsible for IT transformation and integration programs



Key Benefits

- Describes how a business unit uses data, and shows Data Architects what type of data that business unit needs access to and from where they should access it.
- Allows Security Architects to determine critical data – and attempt to secure it.

Data Flow Model



Infrastructure Architecture: Tools and Templates

What is Infrastructure Architecture?



1

Business Architecture

- Defines the business strategy, governance, organization and key business processes.

2

Data Architecture

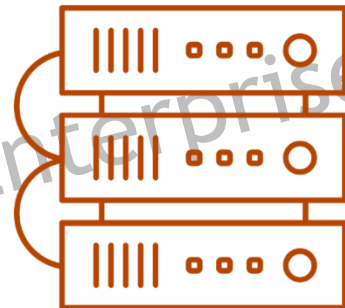
- Describes the structure of an organization's logical and physical data assets and data management resources.



3

Infrastructure Architecture

- Describes the logical software and hardware capabilities that are required to support the deployment of business, data and application services. Includes; IT infrastructure, middleware, networks and communications.



4

Application Architecture

- Provides a blueprint for the individual applications systems to be deployed, and their interactions, and their relationships to the core business processes of the organization.





Infrastructure Business Requirements View

Infrastructure Business Requirements View

Format Type: Can be a table or PowerPoint diagram



Description

Detailed analysis of how an organization's business units need to be supported by infrastructure in order to fulfil their functions. It is usually designed when an organization or business is first created.



Information Required

To complete this view it is best to develop an entire service catalogue of infrastructure first, then assess what the organization needs to deliver those service. Note most organizations are moving or have moved to cloud based services.



Intended Audience

- Business Analysts
- IT Architects
- Security Architects
- Project Managers responsible for IT transformation and integration programs



Key Benefits

- Ensures the organization only receives exactly what it needs to operate.
- Allows standardisation of infrastructure delivery across the organization in order to optimize efficiency and ultimately reduce cost.

Infrastructure Business Requirements View

	Private Cloud	Public Cloud
	Sits on a private hosted network and server with limited control. Organization has total control of data	Sits on a multi-tenanted server that hosts other organisations data. Data resides on cloud server location which could be anywhere
Software as a Service	▪ CRM ▪ Data Management Software (such as SharePoint) ▪ Hypervisor manager ▪ SharePoint	▪ Office 365 ▪ ERP ▪ ITSM
Platform as a Service	▪ Core Operations Platforms ▪ API Management ▪ Cyber Security ▪ AWS RDS or Azure SQL Database	▪ Middleware Orchestration ▪ API management ▪ Cyber Security
Infrastructure as a Service	▪ Hosted Data Centre ▪ AWS EC2 or Azure Virtual Machines	▪ Bandwidth Upgrade ▪ Desktop / Laptop as a Service

Recommendation:

Public cloud “SaaS” sets the standard for ease of consumption of IT services by providing feature rich software on demand. The entry cost and on demand availability of public cloud SaaS is usually attractive to most organisations.

However, public cloud may not be appropriate in all circumstances when considering privacy, risk, uniqueness and longer term total cost of ownership.

Reference: <https://docs.microsoft.com/en-us/azure/architecture/aws-professional/services#security-identity-and-access>



Infrastructure Logical Model

Infrastructure Logical Model

Format Type: PowerPoint or Visio



Description

An architecture diagram which depicts all of the organization’s IT infrastructure, including connections etc.



Information Required

It is a requirement to have the completed list of all infrastructure services before completing this view. You will also need to hold stakeholder workshops to agree on the current state architecture.



Intended Audience

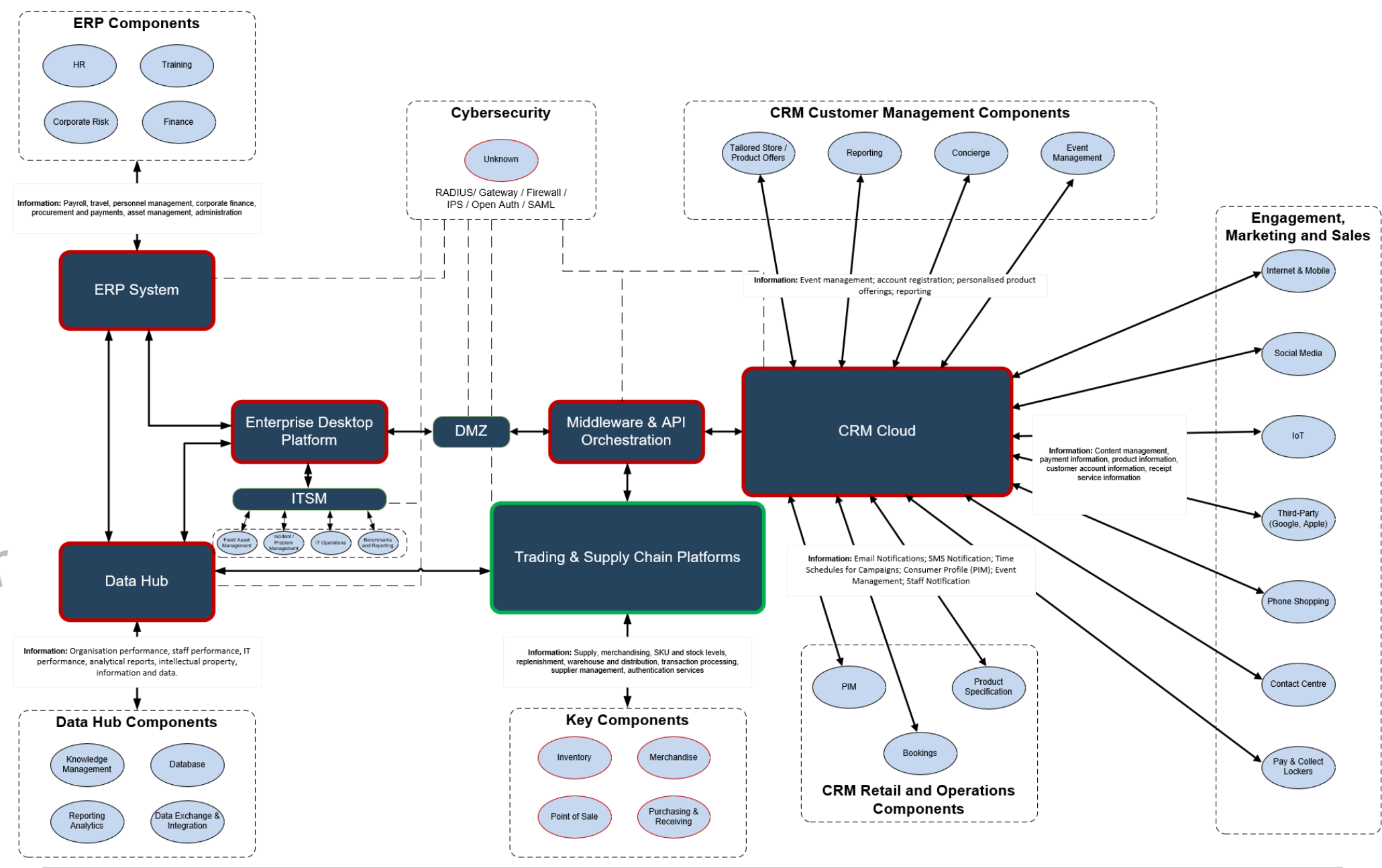
- CIO and Technology Executives
- IT Architects
- Security Architects
- Project Managers responsible for IT transformation and integration programs



Key Benefits

- Provides the most accurate picture of how infrastructure and applications work together to support an application.
- Provides one of the best means for identifying infrastructure gaps.
- One of the best artefacts for identifying security enhancements.

Infrastructure Logical Model





Resource Needs Model

Resource Needs Model

Format Type: Can be depicted in any format



Description

A list of IT resources needed to support an organizations business unit or type of work. It can include anything from people, vendors, cost resources, power needs, ongoing maintenance requirements etc.



Information Required

You firstly need to identify what function or team you are trying to map resources to.
You will also need a list of either infrastructure or applications.



Intended Audience

- CIO and IT Executives
- Business Analysts
- Solution Architects
- Security Architects
- Project Managers responsible for IT transformation and integration programs



Key Benefits

- Provides an excellent visual description of how IT actually supports business teams or business functions.
- Excellent for working with Solution Architects who can help advise on the final list of technology required.

Resource Needs Model

Resource Needs Architecture
(Illustrative Only)



4000 Users

User Access



People Collaboration



Document Collaboration



Foundation





System to Business Cross Reference

System to Business Cross Reference

Format Type: Excel for detailed versions or PowerPoint for summary versions



Description

A simple Enterprise Architecture artefact which shows the business units or teams that connect to the organizations infrastructure.



Information Required

You will need a completed *Business Service View* and *Organisation Model* plus *Service Catalogue* in order to complete this Enterprise Architecture artefact.



Intended Audience

- CIO and IT Executives
- Business Analysts
- Solution Architects
- Security Architects
- Project Managers responsible for IT transformation and integration programs



Key Benefits

- Shows the business interactions across all parts of the organizations IT network.
- Can also help determine waste in IT spend and improve efficiencies.

System to Business Cross Reference

	AWS	Azure	GCP	Oracle Database Server	IBM Server
Sales Team	x				
HR Team			x	x	
IT Team	x	x	x	x	x
Marketing Team		x			
Legal Team	x	x			
Business Operations	x	x	x		
Customer Service Team	x	x	x		
CEO/ Management Team	x				
Finance/ Accounts Team	x				



System to Data Cross Reference

System to Data Cross Reference

Format Type: Excel for detailed versions or PowerPoint for summary versions



Description

A spreadsheet or table which maps infrastructure to data. A primary purpose of this view is to describe how infrastructure is being used to transfer data across an organisation and assess its performance in doing so.



Information Required

It is important to have completed the Data Governance model first before completing this view as you will need to identify the classification and resource needs for data before commencing any type of cross reference.



Intended Audience

- CIO and IT Executives
- Business Analysts
- Solution Architects
- Security Architects
- Project Managers responsible for IT transformation and integration programs



Key Benefits

- Helps Enterprise Architects, IT analysts and Solution Architects determine the best and most appropriate level of infrastructure needed to support the organizations data needs.

System to Data Cross Reference

		Definition	Data Types	Suited Configuration	Infrastructure System
Data Classification Business Criticality ↑	Restricted (Highly Critical)	- Data is classified as Restricted when the unauthorised disclosure, alteration or destruction of that data could cause a significant level of risk to the organisation or its customers (reputation risk). - Systems handling restricted data should have the highest security and resource priority requirements	- Sensitive customer data - Sensitive employee (HR) data - Transaction data - Financial data - PCI data - Regulatory data	- Restricted access should sit on SSD storage (electronic and physical). More expensive option but highly reliable and accessible - Data encryption mandatory - minimum AES 256bit - Secure STMP-TLS relay - No local caching of data - HTTPS only for accessing data through the cloud - Multi-factor authentication required for accessing this data type	- AWS - Azure - GCP - Private on prem server
	Internal (Business / Operations Critical)	Data should be classified as Internal when the unauthorised disclosure, alteration or destruction of that data could result in a moderate level of risk to the organization or its customers	- Marketing data - Asset data - Email data - Vendor and 3rd party data - Non-Restricted customer data - Non-Restricted employee (HR) data - Internal reporting data	- Restricted access (electronic and physical). However, can be HDD storage. - Cheaper option for less critical data. - Data encryption still mandatory - Secure STMP-TLS relay	- Public cloud based applications - Survey Monkey - Workday - Adobe Cloud - Slack
	Public Data / Archiving	Data should be classified as Public when the unauthorised disclosure, alteration or destruction of that data would result in little or no risk to the organization or its customers	- Product data - Marketing data - Archived data - Publication data - Social media data	- Data encryption optional - Secure STMP-TLS relay	- Public cloud based applications - Dropbox - Box - Google Drive



System to Application Cross Reference

System to Application Cross Reference

Format Type: Excel for detailed versions or PowerPoint for summary versions



Description

A simple Enterprise Architecture artefact which shows the applications that connect to the organizations infrastructure.



Information Required

You will need a completed *Service Catalogue* that lists all IT infrastructure and applications in order to complete this Enterprise Architecture artefact.



Intended Audience

- CIO and IT Executives
- Business Analysts
- Solution Architects
- Security Architects
- Project Managers responsible for IT transformation and integration programs



Key Benefits

- Shows the data flows across all parts of the organizations IT network.
- Helps identify security vulnerabilities such as access control.
- Can also help determine waste in IT spend and improve efficiencies.

System to Application Cross Reference

	AWS	Azure	GCP	Oracle Database Server	IBM Server
Salesforce	X				
Office 365	X				
Slack		X			
Survey Monkey	X				
Dropbox				X	X
Canva	X		X		
ServiceNow	X	X	X	X	X
Adobe Cloud	X	X			
Workday	X	X	X		
DocuSign					

Security Architecture Model

Security Architecture Model

Format Type: Excel spreadsheet, PowerPoint of Visio



Description

An architecture artefact that considers all security elements required to protect an infrastructure capability from threat vectors.



Information Required

It is imperative to complete the data architecture views before completing any security architecture for infrastructure capabilities. You will need to know the classification and data types that infrastructure holds or transports.



Intended Audience

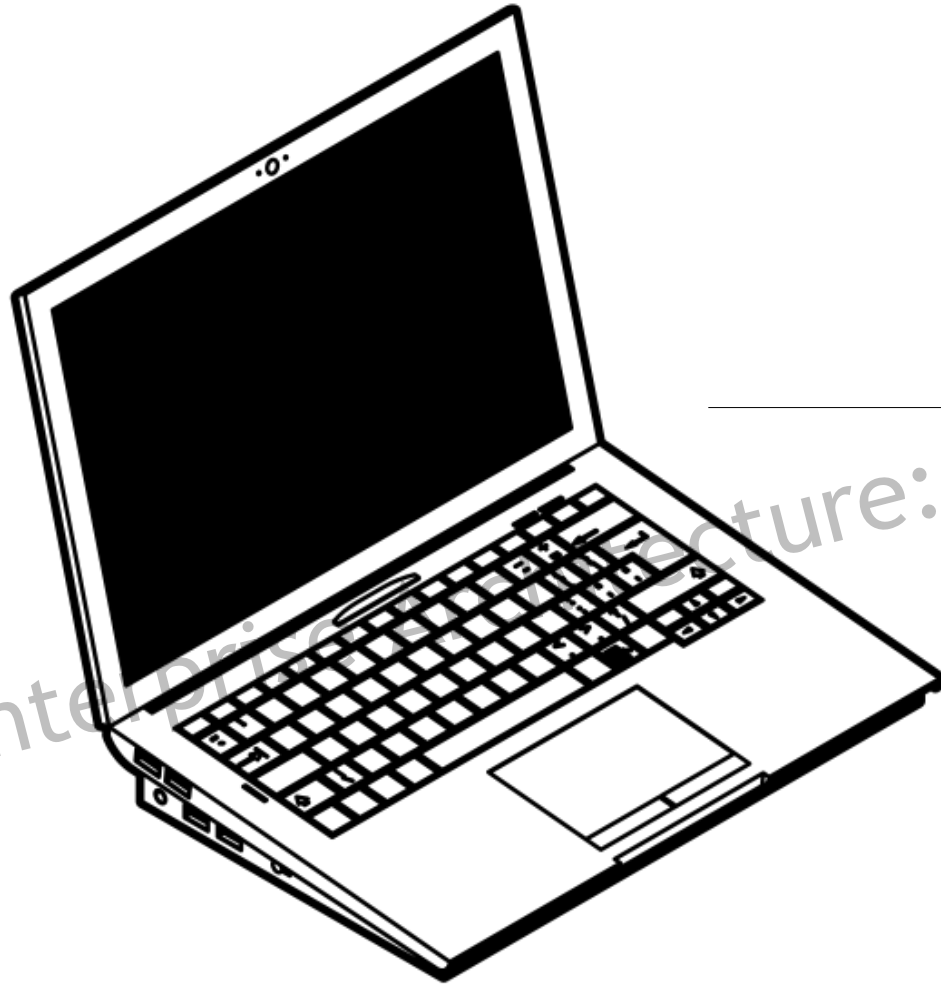
- Solution Architects
- Security Architects
- Project Managers responsible for IT transformation and integration programs



Key Benefits

- Helps secure infrastructure across the IT network.
- Identifies existing security gaps.

Security Architecture Model



- Define laptop vendor
- Define laptop version number
- Define vendor driver/patch number
- Define camera access
- Define USB access
- Define radio and broadcast capability
- Define login protocols
- Define encryption standard
- Define operating system
- Define serial/inventory number
- Define laptop classification
- Define remote user access
- Define destruction date
- Define back-up location
- Confirm security of back-up location
- Define emergency access



Application Architecture: Tools and Templates

What is Application Architecture?



1

Business Architecture

- Defines the business strategy, governance, organization and key business processes.

2

Data Architecture

- Describes the structure of an organization's logical and physical data assets and data management resources.



3

Infrastructure Architecture

- Describes the logical software and hardware capabilities that are required to support the deployment of business, data and application services. Includes; IT infrastructure, middleware, networks and communications.

4

Application Architecture

- Provides a blueprint for the individual applications systems to be deployed, and their interactions, and their relationships to the core business processes of the organization.





Application Business Requirements View

Application Business Requirements View

Format Type: Spreadsheet or table



Description

A artefact that describe the application needs of each business unit with an organization.



Information Required

You will need a completed *Business Service View* and *Organisation Model* plus *Service Catalogue* in order to complete this Enterprise Architecture artefact.



Intended Audience

- CIO and IT Executives
- Business Analysts
- Solution Architects
- Security Architects
- Project Managers responsible for IT transformation and integration programs



Key Benefits

- Shows the application interactions and needs across all parts of the organizations.
- Can also help determine waste in IT spend and improve efficiencies.

Application Business Requirements View

	Human Resources	Business Operations	Sales Team	Finance Team	Marketing Team
Salesforce		X	X	X	X
Office 365	X	X	X	X	X
Slack	X	X	X	X	X
Survey Monkey					X
Dropbox	X				
Canva					X
ServiceNow					
Adobe Cloud					X
Workday	X				
DocuSign	X			X	

| Service Catalogue

Service Catalogue

Format Type: Spreadsheet but can also be built in a formal IT service database such as a Configuration Management Database (CMDB)



Description

A complete detailed list of all applications held and owned by an organisation. It can also include IT infrastructure.



Information Required

You will need to work with the IT team to establish this list if one does not already exist.



Intended Audience

- CIO and IT Executives
- Business Analysts
- Solution Architects
- Security Architects
- Project Managers responsible for IT transformation and integration programs



Key Benefits

- Describes the entire application inventory for the organisation in an easy to manage format.
- Identifies key information such as vendor, application owner, application classification, application version, description of usage, patch/version number, licence information and decommission date.

Service Catalogue

Service Catalogue.xlsx - Excel

FileHomeInsertPage LayoutFormulasDataReviewViewHelpTell me what you want to do

SpellingThesaurusWorkbook Statistics

Check Accessibility

Translate

New CommentDelete PreviousNext

Show/Hide CommentShow All Comments

Protect SheetProtect WorkbookAllow Edit RangesUnshare Workbook

Hide Ink

ProofingAccessibilityLanguageCommentsProtectInk

G370

	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3	Status	Architect	Business Value Chain / Function	IT Capability	IT Common Name	Business Common Name	Application Name	Description	Application Type	Vendor	Core System
348	Live	Bob Smith	Loan Sales	Digital	Afterpay	Afterpay	Afterpay	A payment services method that allows customers to pay for good after they have bought them.	Vendor Application	Afterpay	Y
358											
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Application Architecture Framework

Application Architecture Framework

Format Type: PowerPoint or Visio



Description

A framework used to categorise all applications used by an organization into logical groups.



Information Required

- It is recommended to have a complete list of all applications identified in the *Service Catalogue* first before moving to develop this *Application Architecture Framework*.



Intended Audience

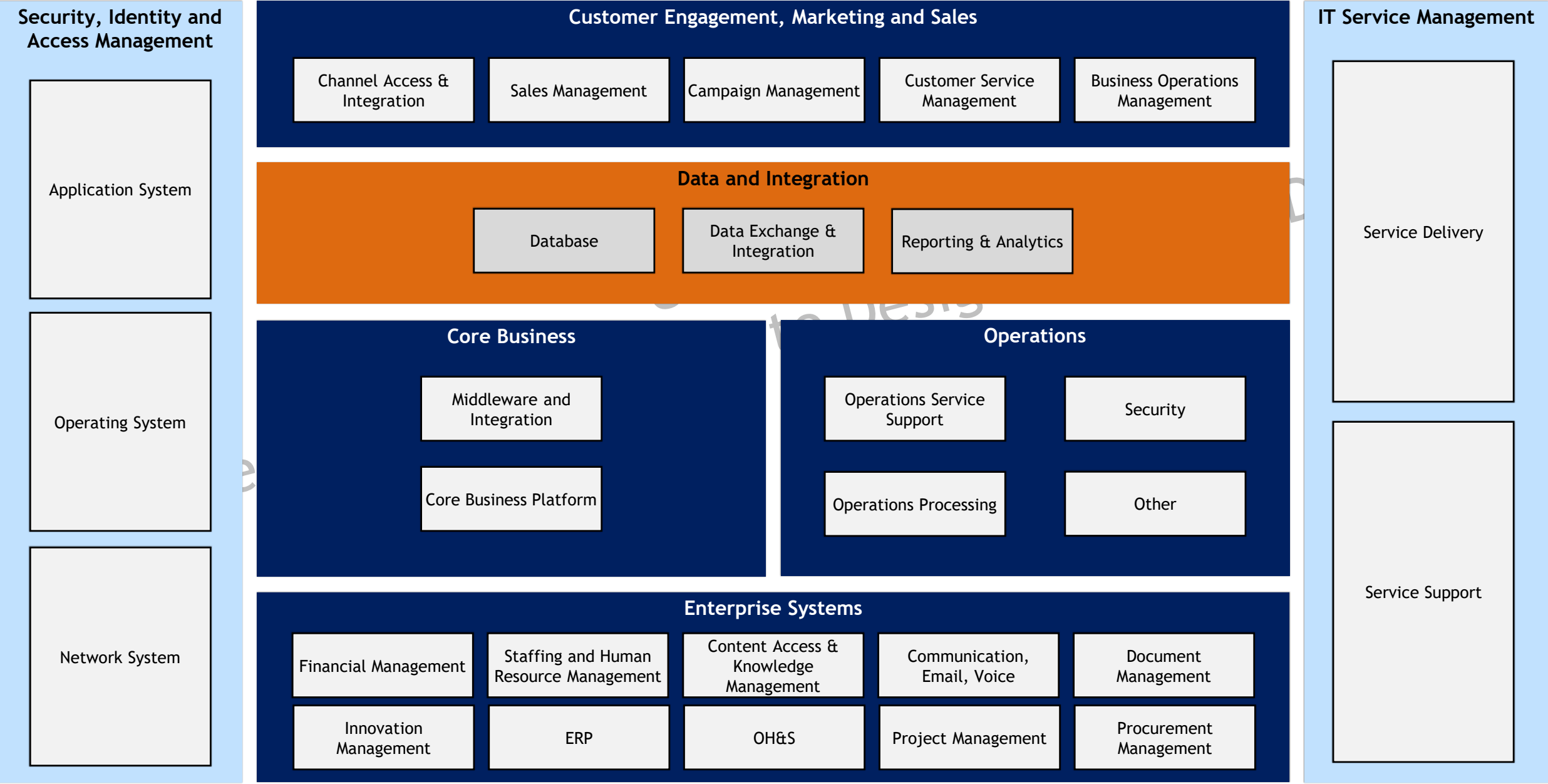
- CIO and Chief Data Officer
- Data Architects and Analysts
- Business Analysts
- Enterprise Architects
- IT Analysts
- Solution Architects
- Project Managers and Consultants



Key Benefits

- Segments applications by customer facing, operations and back of house support applications.
- Allows Enterprise Architects to map applications back to functions and services to ensure the organization is adequately supported by IT.

Application Architecture Framework





Application to Data Cross Reference

Application to Data Cross Reference

Format Type: Spreadsheet or table



Description

A simple architecture view that shows the applications that have an interaction with data or that hold that data.



Information Required

- It is recommended to have a complete list of all applications identified in the *Service Catalogue* first before moving to develop this cross reference.



Intended Audience

- CIO and Chief Data Officer
- Business Analysts
- Enterprise Architects
- IT Analysts
- Solution Architects
- Project Managers and Consultants



Key Benefits

- Can identify which data is crucial for certain business units.
- Identifies if an application is not being fully utilised by the organisation.

Application to Data Cross Reference

	Sales Data	Tax Data	Invoice Data	Marketing Data	Workforce Data
Salesforce	X	X	X		
Office 365	X	X	X	X	X
Slack	X				X
Survey Monkey				X	
Dropbox				X	
Canva				X	
ServiceNow	X	X	X	X	X
Adobe Cloud				X	
Workday					X
DocuSign	X				X



Application to Application Cross Reference

Application to Application Cross Reference

Format Type: Spreadsheet or table



Description

An architecture view that describes all the cross connections and touchpoints between those applications held by the organization. It may also include third-party owned applications.



Information Required

- It is recommended to have a complete list of all applications identified in the *Service Catalogue* first before moving to develop this cross reference.



Intended Audience

- CIO and IT Executives
- Business Analysts
- Enterprise Architects
- IT Analysts
- Solution Architects
- Project Managers and Consultants



Key Benefits

- This is perhaps one of the best architecture views that supports IT integration.
- This view can also show points of failure if an organization relies too heavily on one application and there's no redundancy.

Application to Application Cross Reference

	Salesforce	Office 365	Slack	Survey Monkey	Dropbox	Canva	ServiceNow	Adobe Cloud	Workday	DocuSign
Salesforce	x	x	x		x		x			
Office 365	x	x	x				x			x
Slack	x	x	x		x		x	x		
Survey Monkey										
Dropbox	x		x		x					
Canva						x				
ServiceNow	x	x	x	x	x	x	x	x	x	x
Adobe Cloud			x					x		
Workday							x		x	
DocuSign										x



Security Architecture Model

Security Architecture Model

Format Type: Diagram or model



Description

A security architecture view that describes all the security controls needed to secure an application.



Information Required

- It is recommended to work with the vendor or owner of the application in order to complete this view. Consider also putting this view into a table format if you wish to expand out the detail.



Intended Audience

- Security Architects
- Enterprise Architects
- IT Analysts
- Solution Architects
- Project Managers and Consultants



Key Benefits

- Can quickly identify the vulnerabilities of an application.
- Demonstrates to stakeholders how and why an application needs to be controlled or protected.

Security Architecture Model



chrome

- Define application name
- Define application version number
- Define data transmission security protocol
- Define camera access
- Define USB access
- Define location access
- Define user privacy access and fingerprinting
- Define data store backup and security
- Define clear cache policies
- Define login protocols
- Define encryption standard
- Define operating system needs
- Define remote user access
- Define corporate usage policy
- Define app store policy